

Sébastien Violette

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Highlights of Qualification

- Experienced in full-stack development using TypeScript, SvelteKit, PostgreSQL, and Docker, with a focus on building scalable, research-driven applications and cloud-integrated solutions that meet UX design standards.
- Skilled in automated testing, CI/CD practices, and debugging, utilizing tools like Vitest, Playwright, and Selenium to maintain software reliability and performance.
- Strong understanding of the software development life cycle, object-oriented programming, and Agile methodologies, with hands-on experience in creating responsive, dynamic applications across web and mobile platforms.
- Proficient in building RESTful APIs, integrating third-party services (e.g., Twilio, SendGrid, Stockfish API), and developing solutions across diverse architectures including MVC, SaaS, and middleware-based systems.

Achievements

- Conhacks 2025 Winner** – Hackathon, Conestoga College, Waterloo ON **Feb. 28th, 2025 – Mar. 2nd, 2025**

Projects

- Chess Academia** | **Languages:** Python, JavaScript | **Frameworks:** Flask, FastAPI, Stockfish API, PythonChess, WebSocket
A chess application that allows you to play solo against customizable Stockfish API Elo ratings or against friends online through the generation of unique invitation links and hosting lobbies with the ability to customize the game rules.
- Rejection Therapy** | **Languages:** HTML, CSS, JavaScript | **Frameworks:** Vue.js, React, Express, Node.js, PostgreSQL
(Ongoing) – an interactive web application that humorously delivers automated rejection emails based on user-submitted applications implementing user authentication, form handling, and an email-sending service using Nodemailer and SendGrid.
- EchoMind** | **Languages:** Python, JavaScript | **Frameworks:** Laravel, Next.js, Python, Node.js, PostgreSQL, Twilio
(Ongoing) – Developing an Advanced IVR with Predictive Modeling & NLP, working freelance with peers.

Core Competencies

- Soft Skills:** Cross-functional Collaboration | Mentorship & Networking | Logical Problem-Solving | Efficient | Results-Driven
- Software Development:** Agile Methodologies, Development-Driven Testing, Version Control, Debugging, Data Structures, OOP
- Architecture & Integration:** Scalability, RESTful APIs, Cloud DevOps, SaaS, Web Services, MVC, Middleware Solutions
- Technical Infrastructure:** Performance Optimization, Containerization, Logging & Monitoring, CDN, Networking

Technical Proficiencies

- Programming Languages:** C#, Java, Python, JavaScript, HTML5, CSS, Dart, Flutter, SQL, TypeScript, Svelte, Docker
- Frameworks:** ASP.NET, Azure, Express, React, NodeJS, Figma, Adobe CS, Bootstrap, Tailwind, PostgreSQL, MongoDB

Employment History

Application/Software Developer Co-op | CDS (L4, 4 mo.) – FDS (L5, 8 mo.) | CIBC – Toronto, ON **Sept. 2025 – Aug. 2026**

- Spearheaded the adoption of agentic AI within FINMOD, independently designing and engineering 6 modular AI agents and a dedicated orchestrator, establishing an extensible foundation for future agent development.
- Built automated Jira reporting pipelines for two teams, leveraging GitHub Actions, Azure Data Factory, and Databricks to aggregate team data and distribute weekly/biweekly reports, replacing processes requiring a week+ of manual effort.
- Architected an automated agent governance and changelog system using GitHub Actions that detects agent-file changes across feature branches, analyzes pull requests, and automatically updates changelogs and PRs to improve traceability and auditability.
- Designed a scalable self-improvement framework for 6+ agentic AI systems, collecting hundreds of telemetry signals and using AI models to generate proposed fixes on dedicated feature branches with human review.
- Developed and tested application functionality in COBOL-based mainframe systems, analyzing legacy workflows and contributing to application modernization efforts.

Student Researcher Software Developer Co-op (L2), Conestoga College, Waterloo, ON **May 2025 – Aug. 2025**

- Build and maintain full-stack applications using TypeScript, SvelteKit, PostgreSQL, Docker, and Drizzle ORM to support research-driven software projects, while also developing interactive game experiences using Unity Engine 6 and C#.
- Develop robust automated test suites with Vitest and Playwright, increasing code reliability and reducing bugs in production.
- Contribute to UX prototyping and frontend development with TailwindCSS and accessibility best practices, while streamlining workflows with ESLint, Prettier, and TypeScript.

Education

Advanced Diploma in IT Innovation & Design **Sept. 2023 – Apr. 2027**
Conestoga College, Waterloo, ON | CGPA: 4.0 | Expected Graduation: April 2027

BSc Honours Life Sciences with triple minors **Sept. 2015 – Sept. 2020**
McMaster University, Hamilton, ON | CGPA: 9 – 9.5 (B+)

Certifications & Courses

- Azure Certification (AZ900 & AZ104):** Azure Fundamentals & Administrator **Sept. 2024 – Present**
- Cyber Security Certification** (BrainStation) **Nov. 2024 – Present**